Do Search Agents Know When They Know? Answer-State and Evidence Dynamics in RL-Trained Search Agents

Runguo Li
University of Illinois Urbana-Champaign
Urbana, Illinois, USA
runguol2@illinois.edu

Abstract

Search agents trained with reinforcement learning (RL) answer questions by querying a search engine and reading the returned pages until they decide to stop. Methods that make them stop earlier assume that the agent's own state reveals when its evidence suffices. We test this assumption with an exact diagnostic protocol. At every decision point of a frozen agent, we force the answer it would give if stopped, with or without its own reasoning, measure how retrieved passages change that answer, and replay any stopping policy offline. On seven QA benchmarks, public RL-trained agents often hold the right answer and then lose it: with reasoning allowed, 5–8% of questions are answered correctly at some point but wrongly at the end, two thirds or more of them already correct before any search, and 3–5% beyond what sampling at the final state recovers. Single retrieved passages rarely decide the current answer on their own. The agents' confidence, sample agreement and hidden-state probes separate right from wrong answers only moderately, and none of the stopping methods we evaluate gains significantly. Restating the question after every observation raises accuracy by 0.5–1.7 points (significantly for five of six agents), whereas a placebo reminder does not, and we find no measurable query drift, which points to the agents' reasoning rather than their search. These findings hold under an LLM judge and for a 14B agent. We release SufficiencyBench, 97,841 decision states with oracle labels and an exact replay harness.

CCS Concepts

• **Information systems** → **Information retrieval**; *World Wide Web*; • **Computing methodologies** → *Natural language processing*.

Keywords

Web search agents, retrieval-augmented generation, agentic search, stopping, calibration, benchmark

1 Introduction

Search agents trained with reinforcement learning, such as Search-R1, ZeroSearch, AutoRefine, HiPRAG and their successors, are a widely used recipe for retrieval-augmented question answering on the Web [16, 48, 51, 63]. They interleave reasoning with search calls and decide on their own when to stop and answer. Many recent methods try to make them cheaper. Some shape the reward to penalize tool calls [58, 63, 64], align the decision boundary causally [74], or add dense information-gain rewards [57]; others attach an external controller that stops a frozen agent once the evidence looks sufficient [26, 36, 42]. These methods assume that the agent's state, such as its confidence, its attention to retrieved passages or its hidden representations, reveals when the evidence already suffices, much as language models have been found to "mostly know what they know" in closed-book settings [21]; for search agents, this assumption is rarely tested.

An agent deployed on the Web sends each search call to a search engine, which costs latency and money, and every page it reads can change its answer. We study this stopping decision in a controlled environment that allows exact replay, a Wikipedia snapshot served by the retriever the agents were trained with, and contribute an evaluation protocol, a cross-agent evaluation and a benchmark resource.

Our diagnostic protocol tests this assumption by asking two questions at every decision point of a frozen agent (Figure 1): *what would the agent answer if it stopped now*, and *which retrieved passages actually shape that answer*. Because the agents decode greedily, these counterfactuals are exact, and any stopping policy can be replayed offline on the same states. We apply the protocol to four public RL-trained agents and an untrained reference on 6,125 questions from seven QA benchmarks, all under one retrieval environment, and check the findings with an LLM judge, sampled decoding and a 14B agent.

One in five questions is answered correctly before any search. For 5–8% of questions, the answer forced after the agent's own reasoning is correct at some point but the final answer is wrong, and 3–5% are lost beyond what sampling at the final state recovers, so stopping at the right time would raise accuracy, not only save searches. Most of these answers were known before any search and are usually overwritten by retrieved text [30, 65]; answers found by searching and then lost, as in Figure 1, are rarer (0.5–2.7%). Single passages rarely decide the answer: removing one leaves the current answer unchanged for 70–80% of passages, although removing a whole observation changes it about half the time.

The agents' forced answers are 84–93% confident but 23–30% correct; recalibration repairs this, but confidence, sample agreement, hidden-state probes and supervised heads still separate right from wrong answers only moderately. None of the stopping methods we evaluate gains significantly. Our preference-training recipe at decision points trades abandonment for premature stopping, and DAS [74] leaves abandonment largely in place.

Restating the question after every observation recovers part of the loss, whereas a placebo reminder does not, and we find no measurable drift in the agents' queries, which points to how they reason over what they have read.
Our contributions are as follows.

- We propose a diagnostic protocol that combines forced answers at every decision point, with and without the agent's own reasoning, the causal fate of every retrieved passage

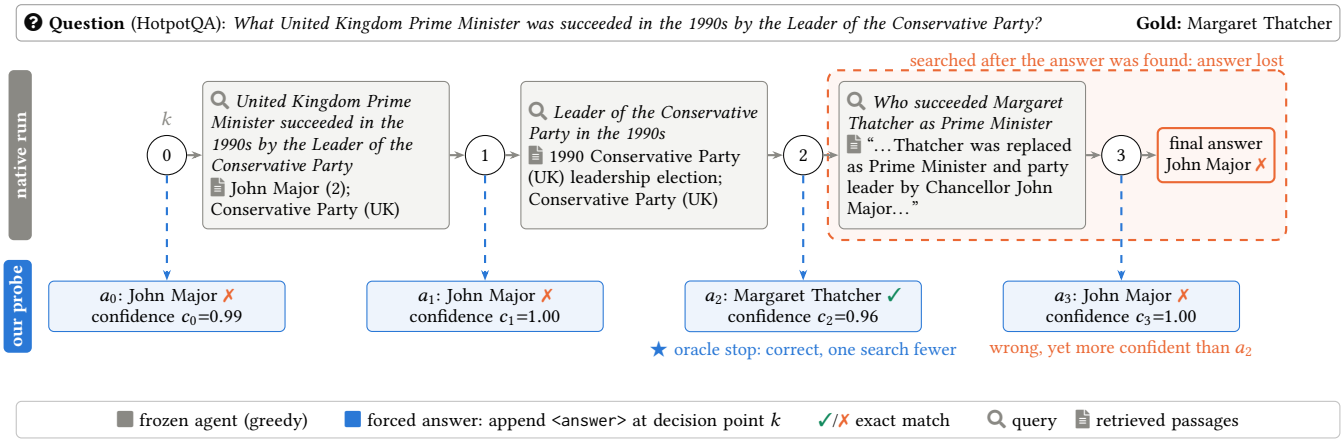


Figure 1: A question that Search-R1-7B answers correctly and then loses. A real trajectory on HotpotQA (top) and the answers we force from the frozen agent at each decision point k (bottom). The agent holds the gold answer after two searches, searches once more and ends with "John Major", while its confidence rises from 0.96 to 1.00. Stopping at k=2 would have been correct and cheaper. Figure 2 shows how we measure this at scale.

under single and joint removal, and exact offline replay with a measured cost model.

- We study answer-state dynamics, evidence use and calibration across four RL-trained agents, an untrained reference and a 14B scale check in one exactly replayable environment.
- We show that re-anchoring, a training-free reminder of the question, raises the accuracy of every agent in the main study, significantly for five of six, and we test it against a placebo reminder and a query-level analysis.
- We build SufficiencyBench, 97,841 decision states with oracle sufficiency labels, baseline signals and an exact replay harness, on which stopping and efficiency methods can be compared without re-running the agents.

Code, trajectories and the benchmark will be released.

2 Related Work

RL-trained search agents. Search-R1 [16] showed that outcome-reward RL teaches LLMs to interleave reasoning with search calls. Follow-up work simulates the search engine (ZeroSearch [51]), adds explicit evidence refinement (AutoRefine [48]), or targets long-horizon web tasks (DeepAgent [27], AEPO [7]). These agents are evaluated by final-answer accuracy, and how their answer state evolves between searches is rarely measured.

Efficient and adaptive search. One line of work decides *whether* to retrieve, before or during generation, from the model's own uncertainty or internal states. Examples include FLARE [14], Self-RAG [2], DRAGIN [50], SeaKR [70], CtrlA [28] and Probing-RAG [3], and Moskvoretskii et al. [33] compare 35 adaptive-retrieval methods systematically. A second line trains search agents to use fewer calls with efficiency-aware or process rewards, as in OTC [58], β-GRPO [64], IKEA [13], HiPRAG [63], SAAS [54], AutoSearch [52] and DAS [74]. IGPO [57] instead uses the per-turn gain in gold-answer probability as a dense reward. A third line attaches an external stopping controller to a frozen agent, as Stop-RAG [36], S2G-RAG [26] and HALT [42] do. DAS [74] is closest to our analysis. It compares factual and counterfactual trajectories at decision points to diagnose over- and under-search, and aligns the agent by preference optimization. We instead probe every decision point of frozen agents without retraining and replay stopping policies exactly. We also find that our preference-training recipe at decision points collapses into a per-turn stopping prior, and that DAS, released or re-trained in our environment, saves some searches but leaves most of the headroom (Section 8).

How LLMs use retrieved context. Sufficient Context [19] separates errors caused by insufficient context from errors in using it. Irrelevant or badly placed context is known to mislead LLMs [29, 47, 72], and when retrieved text contradicts what a model already knows, models often adopt the text, a context–memory knowledge conflict [30, 62, 65, 66]. Attention-based studies show that some heads track context use (Lookback Lens [5], ReDeEP [53], LUMINA [71]) and that such heads can re-rank documents without training (ICR [4], QRHead [75]). Leave-one-out and utility-based evaluation (eRAG [43], UDCG [55]) measures the downstream value of each retrieved passage for single-turn RAG. NeocorRAG [37] reports that higher recall does not translate proportionally into accuracy, and PruneRAG [15] measures how often retrieved gold evidence goes unused. We extend causal, per-document attribution to the multi-turn setting of RL-trained search agents and relate it to their answer dynamics and calibration; many of the answers they lose are lost to such conflicts.

Drift and context interference in search agents. Multi-turn search agents are known to drift from the original question. New evidence can replace the search target or drop its constraints [34], and without an upfront plan, queries drift toward partially relevant documents [39]. Xue et al. [68] find that interference comes mainly from the most recently retrieved documents and train a context

refiner. Test-time contextualization and de-duplication modules improve Search-R1 with an external LLM [46]. In single-turn settings, re-reading the question helps reasoning (RE2 [67]), and regenerating the context around the question reduces distraction (S2A [61]). These remedies are single-turn, or they require training or an additional model. We quantify how often losing sight of the question costs a correct answer and show, with a placebo control, that restating the question after every observation recovers part of it, without measurable query drift.

Calibration after RL. Language models are reasonably calibrated about closed-book answers [21], and sampling-based estimates such as self-consistency and semantic entropy [23, 60] are common uncertainty signals. However, RL post-training is known to collapse policy entropy [6, 7]. Stopping rules decide from the agent's own state whether the evidence suffices, so we measure this effect for search agents under one retrieval environment.

3 Diagnostic Protocol

Agents and environment. We study four publicly released RL-trained search agents and one untrained reference in a single retrieval environment. The RL agents are Search-R1 (Qwen2.5-7B and 3B, PPO) [16], ZeroSearch (Qwen2.5-7B) [51] and AutoRefine (Qwen2.5-7B) [48]. They were trained with policy-gradient RL (e.g., PPO [44], GRPO [45]) on outcome rewards, and AutoRefine adds a retrieval reward (Appendix A). The untrained reference is Qwen2.5-7B-Instruct [40] prompted with the Search-R1 template, and the released Search-R1-14B serves as a scale check. HiPRAG [63] answers in free-form sentences, for which several of our measures are undefined, so Appendix C reports it separately. Every agent keeps its own training-time prompt, observation format, loop protocol and search budget (Appendix A), and decodes greedily with vLLM [25]. Retrieval follows Search-R1, with E5-base [59] over the Wikipedia-2018 dump of [22] (21M passages), exact inner-product search and the top-3 passages per query. We evaluate on NQ [24], TriviaQA [20], PopQA [31], HotpotQA [69], 2WikiMultihopQA [10], MuSiQue [56] and Bamboogle [38], using the test split where available and the dev split otherwise, as released by FlashRAG [17]. We sample 1,000 questions per dataset with a fixed seed (all 125 for Bamboogle), 6,125 questions in total, and a disjoint calibration pool of 100 questions per dataset except Bamboogle (600 in total). Averaged over datasets, our Search-R1-7B run reaches 42.2 EM, compared with 43.1 reported.

Answer-state probing. A trajectory alternates generated text and retrieved observations (Figure 2). Its decision points are the point before any search (k=0) and the point right after each observation ($k \geq 1$ after the k-th). At each decision point k, we append the agent's answer tag (e.g., `<answer>`) to the exact prefix and greedily decode a *forced answer* a_k. Because decoding is greedy and the prefix is identical, a_k is exactly what the agent would output if it were stopped at k. This direct forcing skips the reasoning the agent would write before answering, so we also force the answer after that reasoning: we keep the agent's own text from k up to the point where it would search or answer, which under greedy decoding is exactly its native continuation, and append the answer tag there (*reasoning-allowed* forcing; search is not allowed). At the last decision point, where that continuation ends in an answer, the reasoning-allowed answer equals the native one for 98.7–99.6% of the RL agents' trajectories (direct: 62–84%), a check of the implementation; at earlier points it leads to a search, and forcing remains an intervention. We report both and use direct forcing where a single answer per point is needed, since it is cheaper and gives the same conclusions. The exact match (EM) of a_k against the gold answers gives an oracle label for whether answering now is correct. The geometric mean of the token probabilities of a_k (the exponentiated mean token log-probability) serves as its confidence c_k.

Causal fate of evidence. For every passage d in the k-th observation, we re-probe the forced answer with d replaced by an omission marker and everything else fixed. We label d *necessary* if removing it makes a correct forced answer wrong, and *harmful* if it makes a wrong one correct. It is *influential* if the answer changes without flipping correctness, and *supportive* if the answer is unchanged but its log-probability drops by more than one nat. All other passages have *no marginal effect*.

Exact offline replay of stopping policies. A stopping policy observes the agent at decision points and may stop and answer with a_k. Since stopping only truncates a greedy trajectory, any policy can be replayed exactly from one native run plus the per-point forced answers, without re-running the agent. This lets us sweep thresholds and compare methods on identical states, which we release as SufficiencyBench.

Cost. We report searches, prefill and decode tokens, and latency. We estimate latency as $\alpha \cdot \text{prefill} + \beta \cdot \text{decode} + \gamma \cdot \#\text{searches}$ and measure (α, β, γ) per model size on an A800 GPU (Appendix B). Each controller is charged for every probe it uses.

Statistics and reporting. We test paired accuracy differences between two runs on the same questions with the exact McNemar test [32] ($^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$), and estimate the uncertainty of descriptive statistics with a question-level bootstrap [8] (1,000 resamples). When a comparison is repeated over agents or methods, we also report Holm-adjusted p-values [11]. Accuracy is EM micro-averaged over the 6,125 questions (Search-R1-7B: 42.0), with two exceptions. Stopping methods are replayed on the SufficiencyBench subset of 300 questions per dataset (all of Bamboogle), on which Search-R1-7B reaches 40.2. The paired comparisons of Section 8 score a trajectory that ends without an answer by its last forced answer, which gives Search-R1-7B 42.8 and lowers its abandonment rate (defined in Section 4) from 9.1% to 8.3%.

Robustness checks. Exact match can reject paraphrases, so we relabel all answers with a strict LLM judge (DeepSeek-V4.1-Flash). An answer counts as correct if string matching accepts it, or if the judge accepts it both without and with reasoning mode. We also repeat the pipeline on a second question sample, with top-5 retrieval, with sampled trajectories (temperature 0.7), and for Search-R1-14B (Appendix B). Appendix B gives implementation details for every component.

4 When Is the Answer Already There? (RQ1)

Table 1 shows that for every RL-trained agent the answer is often available long before the agent stops. Between 17.6% and 23.1%

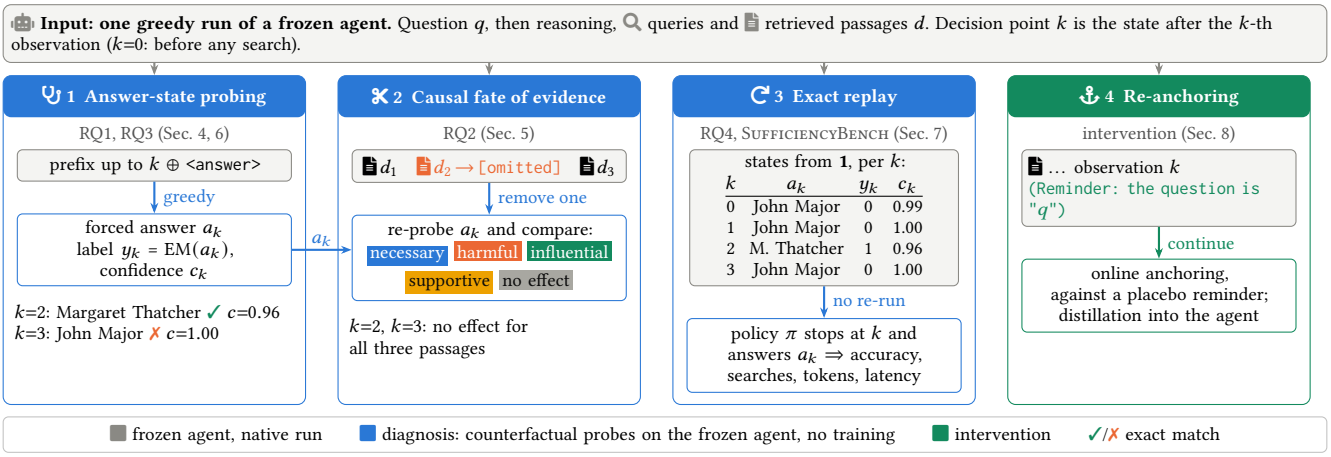


Figure 2: The diagnostic protocol and re-anchoring. From one greedy run of a frozen agent, we (1) force an answer at every decision point and record its correctness and confidence, (2) remove newly retrieved passages, one at a time and jointly, and re-probe to label their causal fate, and (3) replay any stopping policy exactly on the recorded states, which we release as SufficiencyBench. (4) Re-anchoring restates the question after every observation. Example values are from the trajectory in Figure 1.

Table 1: When is the answer already there? Accuracy (EM), searches per question, share of questions whose forced answer is correct before any search (closed-book), at some decision point (reached), and at some point but not at the end (abandoned), also with forced answers that keep the agent's own reasoning (w/ reasoning) and counting only questions for which none of 8 sampled answers at the last decision point is correct (beyond sampling, direct forcing); share of all searches issued after the first correct forced answer, in total (all) and on questions already correct before any search (from closed-book); and the oracle stopping headroom (stop at the first correct point, closed-book allowed), whose accuracy gain equals the abandoned share by construction. The last row is the untrained reference.

					Abandoned			Search after correct		
Agent	Acc.	Search	Closed-book	Reached	all	w/ reasoning	beyond sampling	all	from closed-book	Oracle ΔSearch
Search-R1-7B	42.0	2.42	22.0	49.1	9.1	8.2	5.4	27.3	19.0	-27%
Search-R1-3B	32.8	1.50	17.6	38.6	9.0	7.0	3.9	19.9	16.6	-20%
ZeroSearch-7B	33.8	1.06	20.4	37.0	7.9	5.2	3.6	20.8	20.1	-21%
AutoRefine-7B	43.6	1.81	23.1	48.5	6.8	5.3	3.8	25.0	20.1	-25%
Qwen2.5-7B-Inst.	32.8	1.38	16.0	32.8	6.7	3.2	3.2	15.3	13.5	-15%

of questions are answered correctly before the first search, but the agents still search. Of all search calls, 20–27% are issued after the forced answer is already correct, but most of them (70–97%) come from these closed-book questions, a known form of over-search; searches after an answer that was found by searching are only 1–8% of all searches. We call a question *abandoned* if the agent's forced answer is correct at some decision point but its final answer is wrong, which happens for 6.8–9.1% of questions. When the agent may first write its own reasoning, 5.2–8.2% of questions remain abandoned. Most abandoned questions were correct before any search (56–71%; 67–90% with reasoning), and in 66–76% of these the wrong final answer appears in a later passage: retrieved text overrides a correct parametric answer [66]. Answers found by searching and then lost, as in Figure 1, make up 2.3–4.0% of questions (0.5–2.7% with reasoning). The untrained reference shows the same pattern at a lower level, with 16.0% of questions correct before any search and 6.7% abandoned, but only 3.2% with reasoning (9% of the questions it reaches, against 11–18% for the RL agents).

Taking the best of several answers is optimistic, so we compare the oracle over decision points with best-of-K sampling at the last decision point, with K the question's number of decision points. Sampling recovers 3.5–5.9 points, about half of the oracle's 6.8–9.1. For 3.6–5.4% of questions the agent is correct at some point and none of eight sampled answers at the end is, which we take as a conservative abandonment rate, and 3.3–5.4% if the question must also be abandoned with reasoning. Exact match is strict, so some abandoned questions are surface-form near misses, such as "Armenian" followed by "Armenia" (21–26%); excluding answers that pass a lenient match (token F1≥0.5 or containment), 5.0–7.2% remain (Table 9).

Sources of abandonment. Strict abandonments usually end with an answer taken from later evidence (Table 9). For the four RL

agents, in 54–59% of them the wrong final answer appears verbatim in a passage retrieved *after* the agent was already correct. For the untrained reference, this holds in 41% of cases, and more of its losses (36%) drift without textual support. For Search-R1-7B, a further 24% of strict abandonments end without any answer because the search budget runs out. Many losses are short-lived: with reasoning, only 0.4–2.9% of questions are correct at two consecutive decision points and still lost. An oracle that stops at the first correct decision point would improve accuracy by 6.8–9.1 points while cutting searches by 20–27% (Table 1) and latency by 17–27%, even when charged for its forced answers. These statistics are stable under bootstrap resampling, a second question sample, top-5 retrieval and sampled trajectories (Appendix B), and the LLM judge raises accuracy by 7–12 points but not abandonment (7.0–9.4% vs. 6.7–9.1%).

5 Where Does Retrieved Evidence Go? (RQ2)

Removing a single newly retrieved passage has no marginal effect on the agent's current forced answer for 70–80% of passages across the four RL agents (Table 2). Only 2.6–4.2% of passages are *necessary*, meaning that removing them turns a correct answer wrong, and 1.6–3.0% are *harmful*, meaning that removing them turns a wrong answer right. Gold supporting passages behave similarly, and 69–79% of them also have no marginal effect when retrieved, because the answer was already settled or another passage carries the same fact.

Single removal cannot see redundancy, so we also remove every pair of passages and the whole observation. Removing the whole observation changes the forced answer in 46–64% of observations and turns a correct answer wrong in 9–14%, and for 20–23% of observations only a joint removal changes the answer. Over all eight removal sets, 78–85% of passages have zero Shapley value for the correctness of the current answer, but only 36–58% are also below one nat for its log-probability. These measures concern the current answer only, and a passage can still steer the next query or the final answer.

6 Do Agents Know When They Know? (RQ3)

A stopping rule needs the agent's state to reveal whether its current answer is right, and Table 3 separates two properties of that state, calibration and discrimination. For the four RL agents, forced answers carry a mean confidence of 0.84–0.93 but are correct 23–30% of the time (expected calibration error, ECE, 0.54–0.67 with 15 equal-width bins [9]); Platt scaling fitted on other questions lowers the ECE to 0.05 or less. Discrimination, which a stopping rule needs and post-hoc calibration cannot change, is moderate: the AUC of confidence is 0.73–0.78, and 0.69–0.78 when the agent reasons before answering. Sampling-based uncertainty [23, 60] does not help either: all eight samples coincide at 40–65% of decision points, including 31–57% of those with a wrong answer, and for every RL agent agreement discriminates worse than confidence (AUC 0.66–0.72). Reading the agent's hidden state with a linear probe trained on a disjoint calibration pool gives only a modest gain (AUC 0.74–0.77). The untrained reference is equally miscalibrated (ECE 0.67), which suggests that RL does not create the overconfidence, but its uncertainty is more informative. The untrained model keeps diverse samples, with answer entropy 0.99 against 0.34–0.64 for the RL agents and full collapse at 32% of decision points against 40–65%. Its agreement signal separates right from wrong answers better (AUC 0.82 vs. 0.66–0.72), and so does its hidden-state probe (0.84 vs. 0.74–0.77). Only 52% of its passages have no marginal effect, against 70–80%. With a single untrained reference, this is only suggestive evidence that outcome-reward RL collapses the variability that uncertainty-based stopping would need.

7 How Much Headroom Do Stopping Methods Recover? (RQ4)

We replay each stopping method exactly on SufficiencyBench (Table 4, Figure 3). For all five agents, the label-free rules trace nearly the same accuracy–cost curve as a fixed search budget. None of them improves accuracy by more than 0.4 points, and none of the gains is significant, whereas 13 of the 36 comparisons with the native agent (Table 4 and Search-R1-14B) are significant losses after Holm correction, among them HALT for four agents. The best operating points save 15–26% of searches at a loss of at most half a point for Search-R1-7B, and those of the other agents save at most 14%. The external verifier HALT, attached to the same frozen trajectories with its released adapter, loses 1.2–3.0 points while saving at most 13% of searches; because our setting differs from the original (open-corpus retrieval, the agent's forced answer as the candidate; Appendix B), this does not test HALT as published. The oracle, in contrast, gains 6.6–9.4 points while saving 16–27% of searches (3.8–6.4 for the RL agents without closed-book stops). Every rule's 95% interval ends at or below +1.3 points and the oracle's starts above +5.4, and thresholds tuned per agent on the dev split give −1.3 to +0.3 (Appendix B). With reasoning-allowed forced answers, it gains 4.9–9.0 points for the RL agents and the best rule still at most 0.4. Even with the best threshold in hindsight, choosing an answer among the forced answers and the final one recovers at most 1.0 of the 7–9 points for the RL agents and 1.4 of 6.7 for the untrained one.

Scale. Search-R1-14B is more accurate than Search-R1-7B (45.1 vs. 42.0 EM) but repeats every pattern (Table 7): 26% of its questions are correct before any search, 7.4% are abandoned, its ECE is 0.62 and its probe reaches AUC 0.79. Every stopping method of Table 4 loses 1.2–4.5 points and no threshold that saves searches gains accuracy, while the oracle gains 8.2 points with 33% fewer searches on the replay subset (7.4 on all test questions).

Supervised baselines. Light supervised baselines do no better than the label-free rules (Table 10 in Appendix B). A *learned stopping head*, trained on the calibration pool, estimates whether answering now is correct (p_{now}) and whether the agent's continuation will be (p_{cont}), and stops once $p_{\text{now}} - p_{\text{cont}} \geq \delta$, a one-step value comparison in the spirit of value-based stopping [36], with δ chosen by Learn-then-Test risk control [1]. Its heads reach AUC 0.78–0.86 for p_{now}; under risk control it never stops early, and the best threshold in hindsight gains at most 0.4 points. Learned selection of an answer among the forced answers recovers 0.7–0.8 points on the two strongest agents only. Trained on Search-R1-7B's own rollouts on 20K training questions (63K states), gradient-boosted trees on cheap features reach AUC 0.78 on the test states, but stopping with

Table 2: Causal fate of newly retrieved passages (300 questions per dataset). Single removal: removing one passage of the newest observation makes a correct forced answer wrong (necessary) or a wrong one correct (harmful), changes it without flipping correctness (influential), only lowers its log-probability by more than one nat (supportive), or has no marginal effect; gold supporting passages with no marginal effect (multi-hop datasets). Joint removal: removing the whole observation changes the forced answer, or turns a correct one wrong; observations where only a joint removal changes it; passages with zero Shapley value for correctness over all 2^3 removals, and zero for correctness with less than one nat for the answer's log-probability (inert). The last row is the untrained reference.

		Single removal (%)						Whole observation (%)			Shapley (%)	
Agent	Passages	Nec.	Harm.	Infl.	Supp.	No effect	Gold, no eff.	Changes	Necessary	Only jointly	Zero	Inert
Search-R1-7B	14,043	2.6	1.7	15.2	0.4	80.1	78.4	46.5	9.4	19.9	84.7	57.2
Search-R1-3B	8,559	4.2	2.7	21.6	0.8	70.7	70.4	59.9	12.2	19.8	79.4	49.4
ZeroSearch-7B	6,024	3.7	3.0	22.4	0.7	70.2	69.4	64.2	12.3	21.0	78.2	36.1
AutoRefine-7B	9,870	3.6	1.6	14.5	0.3	79.9	78.5	52.4	13.7	22.5	82.2	57.8
Qwen2.5-7B-Inst.	7,743	3.1	2.3	41.4	1.0	52.2	53.0	74.7	9.0	17.0	83.1	35.9

Table 3: Do agents know when they know? Over all decision points: accuracy of the forced answer, mean confidence, calibration error (ECE) raw and after Platt scaling (cross-fitted over questions), AUC of confidence for direct forced answers and for forced answers that keep the agent's reasoning, AUC of agreement among 8 samples, share of points where all 8 samples coincide (collapse) overall and among wrong answers, and AUC of a linear hidden-state probe trained on the disjoint calibration pool. The last row is the untrained reference.

			ECE		AUC_{conf}					
Agent	Acc.	Conf.	raw	Platt	direct	w/ reas.	AUC_{agree}	Collapse	Collapse|wrong	AUC_{probe}
Search-R1-7B	28.2	93.2	0.649	0.036	0.727	0.726	0.656	65.1	56.7	0.767
Search-R1-3B	23.4	86.1	0.627	0.048	0.755	0.731	0.720	39.8	31.3	0.765
ZeroSearch-7B	24.6	91.2	0.665	0.046	0.729	0.686	0.684	55.7	47.5	0.741
AutoRefine-7B	29.5	83.9	0.544	0.049	0.783	0.784	0.719	48.8	37.6	0.773
Qwen2.5-7B-Inst.	18.0	85.1	0.671	0.029	0.791	0.798	0.816	31.8	22.9	0.838

Table 4: SufficiencyBench: stopping methods replayed exactly on frozen trajectories (ΔEM / Δsearches in % vs. the native agent, 300 questions per dataset). Budget-2: at most two searches; Conf.+stable: stop when confidence ≥0.95 and the forced answer equals the previous one; Agree+stable: all 8 samples agree and stable; Probe: calibration-trained hidden-state probe ≥0.5; $Probe_{cb}$: probe ≥0.8, also allowed to answer before the first search (thresholds from a pilot on Search-R1-7B); HALT: external LoRA verifier on hop claims [42]; $Oracle_{cb}$: stop at the first correct point, closed-book allowed. Paired exact McNemar test vs. native, Holm-adjusted over the 36 comparisons of this table and Search-R1-14B: $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$ (no gain is significant before or after the adjustment). The last row is the untrained reference.

Agent	Native EM	Budget-2	Conf.+stable	Agree+stable	Probe	$Probe_{cb}$	HALT	$Oracle_{cb}$
Search-R1-7B	40.2	−0.5 / -26	−0.4 / -20	−0.3 / -20	+0.3 / -15	−0.3 / -15	-1.7^{**} / -13	$+9.4^{*}$ / -27
Search-R1-3B	32.3	+0.0 / -3	−0.6 / -4	−0.4 / -5	−1.3 / -5	−0.4 / -6	−1.2 / -4	+8.7 / -20
ZeroSearch-7B	32.4	-1.6^{**} / -0	-1.3^{*} / -3	-1.2^{*} / -3	−1.0 / -2	−1.3 / -10	−1.2 / -2	+8.2 / -21
AutoRefine-7B	43.3	−1.4 / -7	−0.7 / -10	−0.6 / -9	+0.2 / -9	−1.4 / -14	-2.5^{***} / -7	+7.0 / -26
Qwen2.5-7B-Inst.	31.8	-1.7^{**} / -12	+0.1 / -1	−0.2 / -2	-1.7^{***} / -2	−0.5 / -6	-3.0^{***} / -5	+6.6 / -16

the threshold chosen on held-out training questions still leaves accuracy unchanged (+0.0 points, 0.3% fewer searches).

8 Re-anchoring: Keeping the Question in View

In RQ1, most strict abandonments ended with an answer taken from a passage retrieved after the agent was already correct, and in Figure 1 the agent ends up answering a question adjacent to the one it was asked. The agent's queries may drift away from the question, or the question may fade as the agent reasons over what it has read. A reminder of the question should help in both cases, but only query drift predicts that the queries of abandoned questions move away from the question and that the reminder changes them.

Online anchoring. *Online anchoring* restates the question after every observation with the line `(Reminder: the question is`

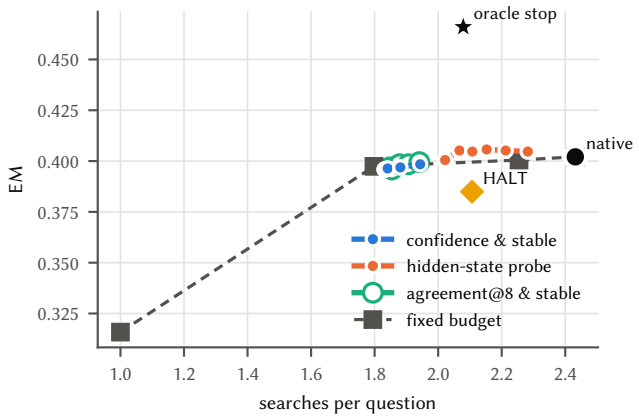


Figure 3: Accuracy–cost frontier of stopping methods for Search-R1-7B on SufficiencyBench. Each curve sweeps the threshold of one signal; "stable" requires the same forced answer at the previous decision point. Fixed budget caps the number of searches, HALT is the external verifier, and the oracle stops at the first correct decision point.

Table 5: Online anchoring on the 6,125 test questions: change in accuracy, searches and latency vs. the native agent, and accuracy change of a placebo reminder without the question and of anchoring vs. placebo. Base: native accuracy, scoring a trajectory that ends without an answer by its last forced answer (as in every run here). Paired exact McNemar test: $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$. Latency uses the measured cost model (7B coefficients for Search-R1-14B). The last row is the untrained reference.

		Online anchoring				Online
Agent	Base	Δacc	Δsrch	Δlat	Placebo	−placebo
Search-R1-7B	42.8	+1.7***	+5%	+6%	+0.3	+1.4***
Search-R1-3B	32.9	+1.6***	-3%	-3%	+0.6	+1.1*
ZeroSearch-7B	33.8	+0.5	-2%	-5%	−0.3	+0.8*
AutoRefine-7B	44.0	+0.9**	-3%	-6%	+0.3	+0.6*
Search-R1-14B	45.6	+1.2***	+0%	+3%	+0.2	+1.0**
Qwen2.5-7B-Inst.	33.0	+1.6***	+4%	+3%	−1.3**	+2.9***

". . . ") and lets the agent continue as usual (placement in Appendix B). The reminder itself is close to re-reading the question in single-turn prompting (RE2 [67]); we use it as a probe of the mechanism behind abandonment rather than as a new method. It raises the accuracy of every agent by 0.5–1.7 points, significantly for all but ZeroSearch also after Holm correction over the six agents (Table 5), and changes searches by −3% to +5% and latency by −6% to +6%. The reminder fixes 1.2–1.7 times as many questions as it breaks (Search-R1-7B: 258 vs. 155). Under the LLM judge, the four RL agents still gain significantly (+0.7 to +2.0) but the untrained reference only +0.6 (p=0.19), so part of its exact-match gain may be a change of answer form (Table 8). The gain grows with the length of the native trajectory (Search-R1-7B: +0.8 points for one search, +4.4 for four), and it holds on questions answered wrongly before any search (+0.9 to +1.7 for all agents but ZeroSearch, $p\leq0.006$).

Table 6: Training the agent. Self-training and anchor distillation fine-tune the agent (LoRA) on its own correct trajectories sampled without or with the reminder, which distillation removes from the training context; no reminder is used at test time. DPO and DAS train on decision-point preference pairs [74]. Acc: native accuracy and change vs. native (paired exact McNemar test, $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$); Reach: correct at some decision point; Aband.: reached but wrong at the end (%); all with the last-point fallback.

	Search-R1-7B			Search-R1-3B		
	Acc	Reach	Aband.	Acc	Reach	Aband.
Native	42.8	49.1	8.3	32.9	38.6	8.9
Self-training	+0.0	49.5	8.6	+2.0***	39.6	8.2
Online anchoring	+1.7***	50.0	6.9	+1.6***	39.3	7.8
Anchor distillation	+1.0**	49.5	7.6	+2.4***	39.6	7.9
DPO	−18.5***	23.1	2.4	−1.2*	36.0	4.8
DAS, released	−0.1	49.1	8.2	–	–	–
DAS, re-trained	+0.3	48.8	7.9	–	–	–

Placebo control. A placebo reminder of the same form and position without the question does not help (−1.3 to +0.6 points, never significantly positive). The question reminder outperforms it for every agent (+0.6 to +2.9, Holm-adjusted $p\leq0.041$), so the gain comes from the question itself. Abandoned questions flip most easily under either reminder (41–60% of the reminder's fixes, 41–56% of the placebo's), and the reminder fixes 1.4–2.3 times as many as the placebo among abandoned and other wrong questions alike, so its advantage is not specific to abandonment. Restating the question only right before the final answer is not enough (−2.0 to +1.2 points).

No measurable query drift. The similarity of the agents' queries to the question falls over the rounds (Search-R1-7B: 0.95 to 0.85 from the first to the fourth query). After the first correct point, however, the queries of abandoned and retained questions differ by only −0.006 to +0.002 within the same dataset and round, and every 95% interval covers zero. An embedding-free measure, the share of the question's content words that a query keeps, agrees (−0.012 to +0.003, all intervals covering zero; for Search-R1-14B, abandoned queries keep slightly more of the question). Anchoring also leaves query similarity unchanged (Search-R1-7B: 0.92 with and without the reminder). Neither measure detects a changed relation, such as Figure 1's query for Thatcher's successor, so we conclude only that the queries show no measurable drift. Appendix B gives two further uses of the reminder.

Training the agent. To distill online anchoring into the agent, we sample four trajectories with the reminder for each of 20K NQ and HotpotQA training questions, keep the correct one with the fewest searches, remove the reminder and fine-tune the agent on its own tokens with LoRA [12] (Appendix B), a form of context distillation [49] with rejection-sampling self-training [73]. A *self-training* control applies the same recipe to trajectories sampled without

the reminder. For Search-R1-7B, the distilled agent gains 1.0 points with no reminder at test time and beats the control by 0.95 points (p=0.004), by lowering abandonment at unchanged reach (Table 6). For Search-R1-3B, self-training alone gains 2.0 points, and the reminder adds 0.4 (not significant). Training against abandonment directly failed: LoRA DPO [35, 41] on step-level pairs that prefer answering at the first correct point of an abandoned trajectory, and searching at a wrong point of a successful one, lowers abandonment (Search-R1-7B: 8.3% to 2.4%) but teaches premature stopping (−18.5 points for Search-R1-7B, −1.2 for 3B). With held-out preference accuracy of 61–84% and one seed per variant, we read this as a failed recipe, not as evidence that preference training cannot work. DAS [74], trained on counterfactual decision-point pairs, changes Search-R1-7B by −0.1 points with its released adapter and by +0.3 when re-trained in our environment (neither significant), and online anchoring still adds 1.2–1.4 points on top of either (p<0.001).

9 SufficiencyBench

We release the decision states behind RQ1–RQ4 as SufficiencyBench, which contains 97,841 decision points on 6,125 questions from six agents: the four RL agents, HiPRAG and the untrained reference. Every point records the question, gold answers, query and retrieved passages, the forced answers with and without the agent's reasoning and their oracle labels (string match and LLM judge), the native continuation's outcome and token cost, and the baseline signals studied here (Appendix B). The evaluator replays a stop/continue policy in order, charges every forced answer it reads to its decode cost, and reports accuracy, searches, tokens and latency. Under greedy decoding, this replay equals re-running the agent under the policy, and the evaluator reproduces the native, fixed-budget and oracle numbers of this paper exactly. We fix a split: methods are fit on the calibration pool (train), may be tuned on the 4,200 test-pool questions outside the replay subset (dev), and are evaluated once on the replay subset of 1,925 questions (test). Replay covers policies that truncate the native trajectory; interventions such as re-anchoring need new rollouts, and we release our online runs (native, anchored and placebo) for comparison. New stopping methods can be compared on identical states in seconds on a CPU, and the released code records these states for new agents.

10 Discussion

Stopping. Most lost answers were known before the first search, and later searches often bring in the passage whose answer the agent ends with, so deciding whether and when to search can raise accuracy as well as save cost. Recalibration cannot add the discrimination that the agents' near-deterministic confidence lacks, so we suggest training agents or light heads on the oracle labels of SufficiencyBench to predict *answer correctness at the current point*, and evaluating them by exact replay.

Training. A one-line reminder, which is not a stopping rule, recovers up to a quarter of the headroom. Preference optimization at decision points taught premature stopping or changed little, perhaps because without a reliable correctness signal (RQ3) the agent cannot tell which stops to prefer; the reminder could instead enter RL rollouts.

Retrieval. Removing a single retrieved passage rarely changes the current answer, a few percent of passages mislead, and neither relevance rank nor attention separates them well (Appendix B). Whole observations matter more, so pruning should remove only passages that have no effect also when removed together with others; detecting passages that would overturn an already correct answer is a second target.

11 Conclusion

RL-trained search agents often hold the right answer before they stop, usually one they knew before searching, and part of this loss remains when they reason before answering and after accounting for sampling. Single retrieved passages rarely decide that answer on their own. Their confidence and internal state reveal only moderately whether they are done, and the stopping signals we test recover little of the oracle's gain in accuracy and cost. Restating the question recovers part of it, the placebo control shows that the question's content matters, and we find no measurable query drift. SufficiencyBench lets new stopping methods be tested cheaply on identical states.

Limitations. Forced answers cannot show the effect of a search the agent did not make, and before the last point they cut reasoning that leads to a search; removal analyses concern the current answer, and our query measures cannot detect a changed relation. The LLM judge is imperfect (Appendix B), and exact replay requires greedy decoding. We study short-answer QA over Wikipedia; deep-research agents on the live Web may behave differently. All agents build on Qwen2.5, the scale check and the untrained reference are single models, and our HiPRAG reproduction falls short of its reported accuracy (Appendix C). Re-anchoring and distillation used one wording and one recipe each, DPO three variants on two agents, and each training run a single seed.

Ethical considerations. We use only public QA benchmarks and Wikipedia, with no human participants or user data.

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A Agents and Protocols

All checkpoints are the public releases on the Hugging Face Hub (PeterJinGo/SearchR1-nq_hotpotqa_train-qwen2.5-{7b,3b}-em-ppo and -14b-em-ppo-v0.3, Alibaba-NLP/ZeroSearch_google_V1_Qwen2.5_7B, yrshi/AutoRefine-Qwen2.5-7B-Base, qualidea1217/Qwen2.5-3B-Instruct-GRPO-HiPRAG, and Qwen/Qwen2.5-7B-Instruct). Search-R1 (7B, 3B and 14B; outcome EM reward, PPO), ZeroSearch-7B (outcome reward, simulated search during training), AutoRefine-7B (outcome and retrieval rewards) and the untrained Qwen2.5-7B-Instruct run four search rounds of 500 new tokens each and one final round without retrieval, read passages inside `<information>` tags (AutoRefine: `<documents>`), and give forced answers of at most 32 tokens after `<answer>`, scored by EM. HiPRAG-3B (outcome and process rewards, GRPO) runs ten rounds of 1,024 tokens with `<context>` observations and no final round; its forced answers of at most 128 tokens follow `</conclusion></step></think><answer>` and are scored by cover EM (CEM). Search-R1-style agents generate until `</search>` or `</answer>`, receive passages as `Doc` i`(Title:` t`) text`, and get Search-R1's "invalid action" message otherwise; HiPRAG appends `<conclusion>` to an output with neither a search nor an answer that did not end with EOS.

B Implementation Details

Retrieval. Our flat inner-product search in PyTorch uses Search-R1's fp16 vectors and query encoding; on 40 queries, its top-1 passage agrees with faiss [18] CPU search in 97.5% of cases (fp16 ties otherwise).

Forced answers and sampling. Agreement counts how many of K=8 samples (temperature 0.7, top-p 0.95) equal the greedy forced answer after SQuAD-style normalization. A reasoning-allowed forced answer appends the answer suffix after the agent's native text up to its next `<search>` or `<answer>` tag; its confidence covers the answer tokens only, and a replayed stop also pays for the reasoning. Best-of-N counts a question as recovered if any of the first N of the eight samples at the last decision point is correct. Post-hoc calibration fits Platt scaling (on the confidence logit) or isotonic regression with 5-fold cross-fitting grouped by question; isotonic regression gives ECE 0.004–0.011.

Causal fate and joint removal. On the first 300 questions per dataset, each passage of the newest observation is replaced by `[omitted]` (title kept), the forced answer is re-decoded, and the log-probability of the original answer is re-scored in the spelling the agent prefers. We also remove each pair of the three passages and all of them; with the single removals, this gives the forced answer for all eight subsets of kept passages and exact Shapley values for the answer's correctness and for the log-probability of the original answer.

Hidden-state probe and attention. We take the residual stream just before the first answer token at 25, 50, 62.5, 75, 87.5 and 100% depth and fit a standardized logistic regression (C=0.05) on it plus the confidence and a closed-book indicator. The layer is selected by 5-fold cross-validation grouped by question on the calibration pool, and test AUC is reported once. For the attention signals, forward hooks reduce attention to the per-head mass that forced-answer tokens put on each passage. Heads are selected by AUC on one third of the questions and evaluated on the rest, and the linear variant fits a logistic regression (C=0.01) on all heads. For Search-R1-7B, attention predicts a passage's causal usefulness only modestly (AUC 0.72 for the best 16 heads, 0.77 for the linear variant, 0.58 for retrieval rank).

HALT. With the released LoRA verifier and HALT's prompts, each question is decomposed into two hop claims, each (claim, sentence) pair is labeled Match, Partial or Null, and the agent stops once every claim is matched. Unlike the original setup, the verifier

Table 7: Robustness: main statistics under a second question sample, top-5 retrieval and sampled search trajectories (temperature 0.7; forced answers stay greedy), and for Search-R1-14B (micro-averaged EM; causal fate on 300 questions per dataset, 200 at temperature 0.7).

Agent	Setting	EM	Search	Closed-book	Aband.	Strict	Search after correct	Oracle ΔSearch	No effect
Search-R1-7B	main (seed 2027, top-3)	42.0	2.42	22.0	9.1	7.0	27.3	-27%	80.1
	resampled (seed 2028)	43.1	2.43	21.5	8.7	6.4	27.1	-27%	78.9
	top-5 passages	42.8	2.37	21.8	8.8	6.4	27.5	-28%	83.7
	temperature 0.7	42.0	2.42	21.8	8.9	6.9	27.1	-27%	78.9
AutoRefine-7B	main (seed 2027, top-3)	43.6	1.81	23.1	6.8	5.0	25.0	-25%	79.9
	resampled (seed 2028)	45.5	1.81	23.9	6.6	4.8	25.5	-26%	78.0
	top-5 passages	44.7	1.76	23.1	6.7	4.8	25.2	-25%	84.3
	temperature 0.7	43.5	1.80	23.1	7.0	5.3	24.9	-25%	79.7
Search-R1-14B	main (seed 2027, top-3)	45.1	2.36	26.4	7.4	5.7	32.4	-32%	80.4

Table 8: Main statistics under string matching (S), the strict LLM judge (F) and the conservative label (B: string match, or both judge modes accept); micro averages over 6,125 questions; online anchoring with the last-point fallback (McNemar, $^{*}p<0.05$, $^{**}p<0.01$, $^{***}p<0.001$). The reasoning-mode judge covers 7,383 of the 8,235 answer pairs the fast judge accepted; the rest count as wrong under B, so B is a lower bound. The last row is the untrained reference.

	Accuracy			Abandoned			Closed-book		Search after corr.		Online anchoring	
Agent	S	F	B	S	F	B	S	B	S	B	S	B
Search-R1-7B	42.0	51.9	50.5	9.1	9.5	9.2	22.0	25.6	27.3	32.2	$+1.7^{***}$	$+1.8^{***}$
Search-R1-3B	32.8	41.2	39.8	9.0	9.8	9.4	17.6	20.4	19.9	23.2	$+1.6^{***}$	$+2.0^{***}$
ZeroSearch-7B	33.8	43.2	41.8	7.9	8.3	8.1	20.4	24.4	20.8	24.3	+0.5	$+0.7^{*}$
AutoRefine-7B	43.6	53.9	52.5	6.8	7.2	7.0	23.1	27.3	25.0	29.6	$+0.9^{**}$	$+0.7^{*}$
Qwen2.5-7B-Inst.	32.8	47.0	44.9	6.7	8.1	7.9	16.0	22.1	15.3	20.2	$+1.6^{***}$	+0.6

Table 9: Abandoned questions (%): rate under EM, strict rate (the final answer also fails token F1≥0.5 and containment), and where the wrong final answer of a strict abandonment appears: in a passage retrieved after the first correct point (later ev.), only earlier (earlier ev.), nowhere (drift), or no final answer. The last row is the untrained reference.

Agent	Rate	Strict	Later ev.	Earlier ev.	Drift	No ans.
Search-R1-7B	9.1	7.0	55.8	9.1	11.2	24.0
Search-R1-3B	9.0	7.2	58.6	18.6	20.2	2.5
ZeroSearch-7B	7.9	5.8	57.6	12.4	29.8	0.3
AutoRefine-7B	6.8	5.0	54.4	14.6	15.9	15.2
Qwen2.5-7B-Inst.	6.7	4.2	40.5	12.4	35.9	11.2

runs in bf16, retrieval is our open-corpus index (HALT's Setting B), and the answer candidate is the agent's forced answer.

Learned stopping head and answer selection. The stopping head uses confidence, stability, step, answer length, distinct answers so far, answer occurrence in the passages, the top retrieval score and a middle-layer state, with logistic regressions (C=0.01) or gradient-boosted trees (depth 3, 200 iterations). The most aggressive margin δ whose replay on cross-fitted calibration predictions does not lose accuracy under a one-sided sign test (α=0.1) is kept (fixed-sequence testing). Learned selection scores every candidate answer with the same tree model (frequency and longest run, first and last step, finality, confidence, occurrence in passages, length) and replaces the final answer only if the best margin passes a threshold chosen the same way. The 20K-question heads of Section 7 drop the hidden state, add a closed-book indicator and answer occurrence in the newest observation, and are trained on 57K states of Search-R1-7B's greedy rollouts on the 20K distillation training questions (trees: 300 iterations), with 6K states of 10% of the questions held out to choose the threshold (validation AUC 0.80 for the trees).

Table 10: Supervised baselines trained on a disjoint calibration pool (3.6K questions for Search-R1-7B and AutoRefine-7B, 600 otherwise). Stopping head: logistic heads estimate p_{now} and p_{cont} (AUCs) and stop when $p_{\text{now}} - p_{\text{cont}} \geq \delta$. Under risk control (Appendix B), it gives +0.0 / +0% for every agent. Best in hindsight: best δ on the test set (Δacc / Δsearches). Selection: a small GBT picks among the forced and final answers (share of headroom). The last row is the untrained reference.

Agent	Head-room	Stopping head AUC now / cont	Stopping head best in hindsight	Selection Δacc
Search-R1-7B	9.1	0.81 / 0.72	+0.4 / -10%	+0.8 (8%)
Search-R1-3B	9.0	0.79 / 0.72	+0.0 / -1%	+0.1 (1%)
ZeroSearch-7B	7.9	0.78 / 0.72	+0.0 / -2%	+0.0 (0%)
AutoRefine-7B	6.8	0.83 / 0.73	+0.3 / -4%	+0.7 (10%)
Qwen2.5-7B-Inst.	6.7	0.86 / 0.72	+0.2 / -1%	+0.0 (0%)

Re-anchoring. The reminder (`Reminder: the question is "`q`"`) follows the closing observation tag, except for AutoRefine and HiPRAG, where it goes just before it (after AutoRefine's `</documents>`, 83% of its trajectories ended without an answer). The placebo (`Reminder: read the information carefully before you continue.`) takes the same position. The *re-anchor vote* forces two answers at the native stop point after the reminder,

from the full context and from the passages alone, and returns the majority with the native answer; it gains 0.6–1.2 points for the RL agents (significantly for all four) at 10–13% extra latency and −0.3 to +0.4 on top of online anchoring. *Anchor consistency* forces a second answer with the reminder at every decision point; an answer that changes under it is correct 5–9% of the time, one that survives it 31–43%, and stopping once it is consistent and stable gains 0.4 points with 21% fewer searches for Search-R1-7B.

Distillation. We sample four trajectories at temperature 1.0 for each of 10K NQ and 10K HotpotQA training questions (from the agents' training splits, disjoint from the test questions), with and without the reminder, and keep the correct one with the fewest searches (7B: 11.7K questions per condition; 3B: 10.2K and 10.1K). LoRA (rank 64, α=128, dropout 0.05, all attention and MLP projections) is trained for two epochs at learning rate 10^{-4} (cosine, 3% warm-up), batch 64 and length 4,096, with the loss on agent tokens only.

Anti-abandonment DPO. Pairs come from forced answers at every decision point of 80K sampled training trajectories per agent; each keeps the agent's reasoning (`<think>. . . </think>`) of its next segment in the prompt and contrasts `<answer>` *a* `</answer>` with the actual action. We train one epoch with LoRA (rank 64), β=0.1, NLL weight 0.2, learning rate 2×10^{-5} and batch 64 (3B: 60), with the adapter-disabled model as reference. Raising β to 0.5, or dropping the agent's reasoning from the stop action, gives the same collapse (7B: −18.0 and −20.3; 3B: −1.3 and −1.2).

DAS. The released DAS adapter was trained on the same Search-R1-7B checkpoint (per its code and a likelihood check on its pairs); to re-train it, we follow its released pipeline and prompts with greedy rollouts on our 20K training questions: answer-only continuations at search steps and hint-guided search-only continuations at wrong answer steps, with token F1≥0.2 counted as correct. Its quality filter yields 11,836 over-search and 6,278 under-search pairs, of which we keep 5,000 of each, and we train LoRA DPO with its settings (rank 64, α=128, β=0.3, learning rate 10^{-6}, three epochs, effective batch 16). The re-trained adapter saves 8% of searches but leaves 8.0 points of headroom.

LLM judge. The 95K distinct (gold set, prediction) pairs of the native, online-anchoring and distillation runs are judged by DeepSeek-V4.1-Flash (about 100K API calls in September 2026; the full prompt will be part of the code release). The prompt lists, with examples, what counts as correct (aliases, spelling variants) and incorrect (related entities, vaguer answers, several options, partial lists, wrong numbers or dates); the modes erred differently, so the label requires both to accept and accepts 6.4% of string-rejected answers (fast mode alone: 8.7%). On 500 string-accepted answers, it agrees on 52 of 53 short answers but on 77% of 447 HiPRAG sentences.

Query similarity. Queries and questions are encoded with the retriever's E5 encoder; the lexical measure is the share of the question's content words (SQuAD-normalized, without stopwords) that the query contains. Queries after the first correct point are compared within (dataset, round) cells and averaged with weights equal to the smaller group, with 300 bootstrap resamples of questions. A third proxy, whether such a query names the answer the agent then holds (as in Figure 1), differs by −1.3 to +3.1 points with mixed signs (−7.0 for Search-R1-14B).

Cost and compute. For the 7B agents, the measured latency coefficients are α=6.9×10^{-5} s per prefill token, β=1.1×10^{-2} s per decode token and γ=0.021 s per search. Diagnosis is inference only (2–4 GPU-hours per agent on one H200 or A800, plus one for HALT); distillation took about 25 A800 GPU-hours, the DPO variants 30, the placebo and anchor-consistency runs 15, and the reasoning-allowed, sampled and joint-removal probes about 3.

SufficiencyBench format. Each record is one decision point k of one question and agent, with the question, gold answers, query and retrieved passage identifiers and titles; the direct and reasoning-allowed forced answers with their lengths and oracle labels (EM or CEM, and LLM judge); the native final answer and its labels; native searches and tokens up to this point and for the episode; the baseline signals (confidence, stability, agreement@8, probe probability, HALT decision where available); and the split (train: calibration pool; dev and test as in Section 9).

Uncertainty and robustness. With 1,000 question-level bootstrap resamples [8], 95% intervals have half-widths of at most 1.3 points for the shares of Table 1 and 1.5 for the no-effect share. Table 7 repeats the full pipeline for Search-R1-7B and AutoRefine-7B on a second question sample (seed 2028), with five passages per search and with search trajectories sampled at temperature 0.7, which change these statistics by at most three points (4.3 for the no-effect share), and gives Search-R1-14B. Sampling changes 67% of the trajectories and 20% of the final answers but no statistic by more than 0.3 points (1.1 for the no-effect share). Intervals in Section 7 use 2,000 paired bootstrap resamples of the 1,925 replay questions (80% power to detect 0.5–1.7 points); the thresholds of Table 4 come from a pilot sweep on Search-R1-7B, and the dev check picks, per agent and rule, the one with the best dev accuracy.

C HiPRAG

HiPRAG-3B [63] (Qwen2.5-3B-Instruct, GRPO with a process reward for efficient search) answers in sentences, scored with CEM, for which string-level answer change, exact sample agreement and the verbatim source test are not meaningful, so we omit them. With its released prompt, loop, limits and retrieval setup, it reaches 48.9 CEM versus 65.4 reported, as 8.5% of its greedy trajectories degenerate without a parsable answer, so its numbers are a lower bound. Like the other agents, it answers 23.2% of its questions correctly before any search, abandons 9.3% and issues 31.8% of its searches after a correct answer. It is less overconfident (ECE 0.36) but discriminates no better (AUC 0.71 for confidence, 0.82 for the probe). As its trajectories often end without an answer, early stopping helps it slightly (+2.0 CEM for a two-search budget, +0.5 for HALT, +8.8 for the oracle), as does choosing among its forced answers (+3.7). Online anchoring and the placebo change its accuracy by 0.0 and +0.2 points, and the LLM judge raises it from 51.1 to 54.7.